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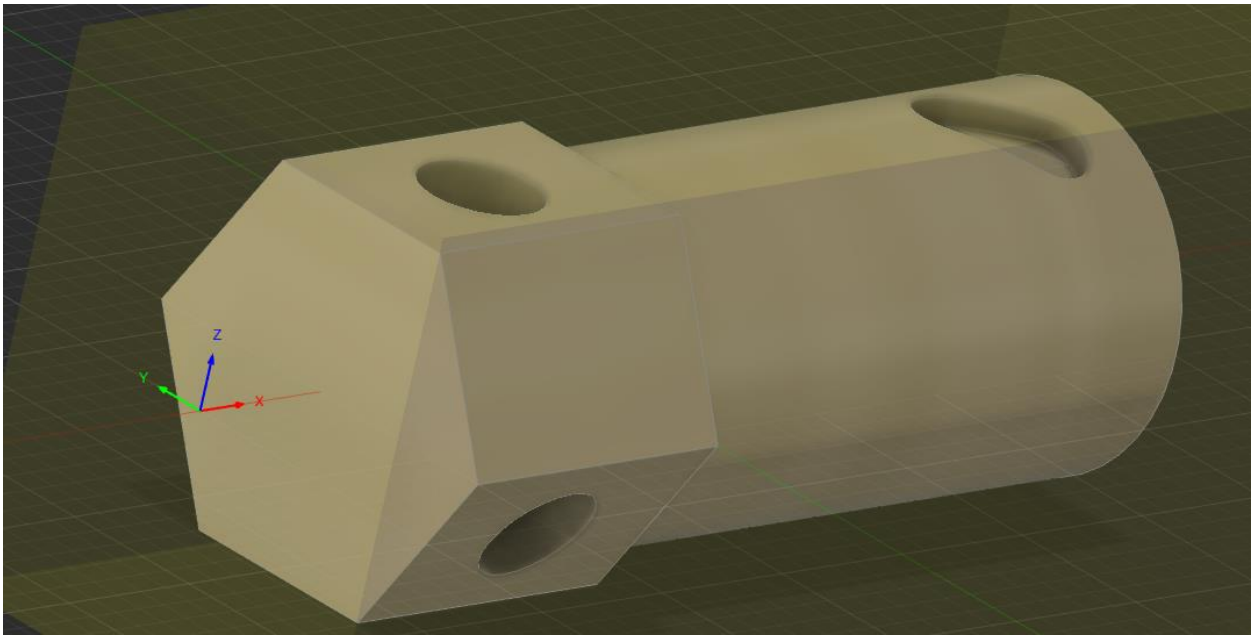
Fusion Setup

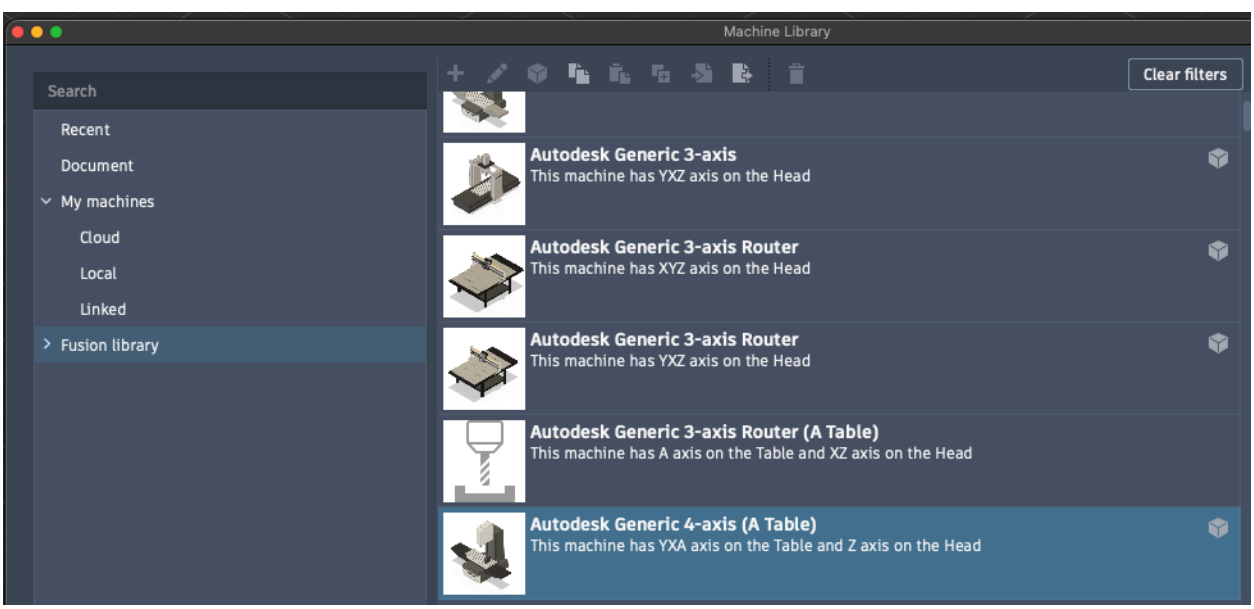
WCS

The work coordinate system X axis, must be along the rotary axis center.



Machine Type

Pick “Autodesk Generic 4-axis (A Table)



Post

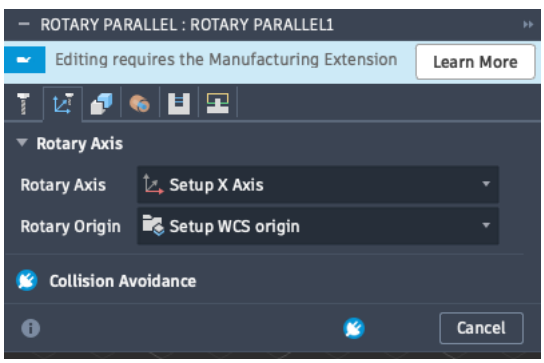
Download the following post:
https://github.com/atwatsoniii/AsmblyLagunaRotary/blob/main/Fusion/Laguna_IQ_Turner_Asmbly.cps

In fusion CAM workspace, go to Manage -> Post Library -> Local. Click the import buttom at the top. Select Laguna_IQ_Turner that was downloaded from above. You can now use this post in you NC programs as normal. Units must be in millimeters, just the like normal Laguna post.

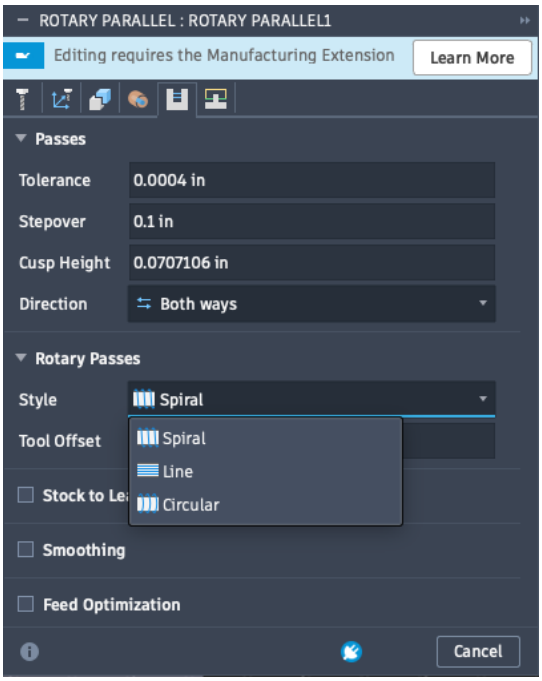
Toolpaths

Multi-Axis Menu

This requires the machining extension. Rotary Parallel is good for a lot of cases. It can do any kind of complex geometry. In the multi-axis tab, make sure that Rotary Axis is “Setup X axis”, and Rotary Origin is “Setup WCS Origin”

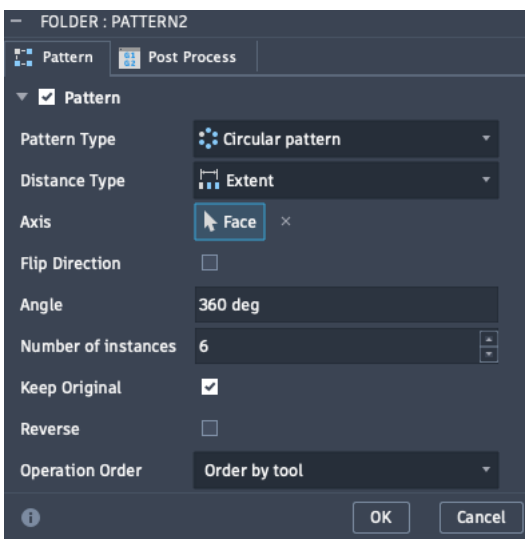


In the passes tab, there are options for how the rotary passes are done:

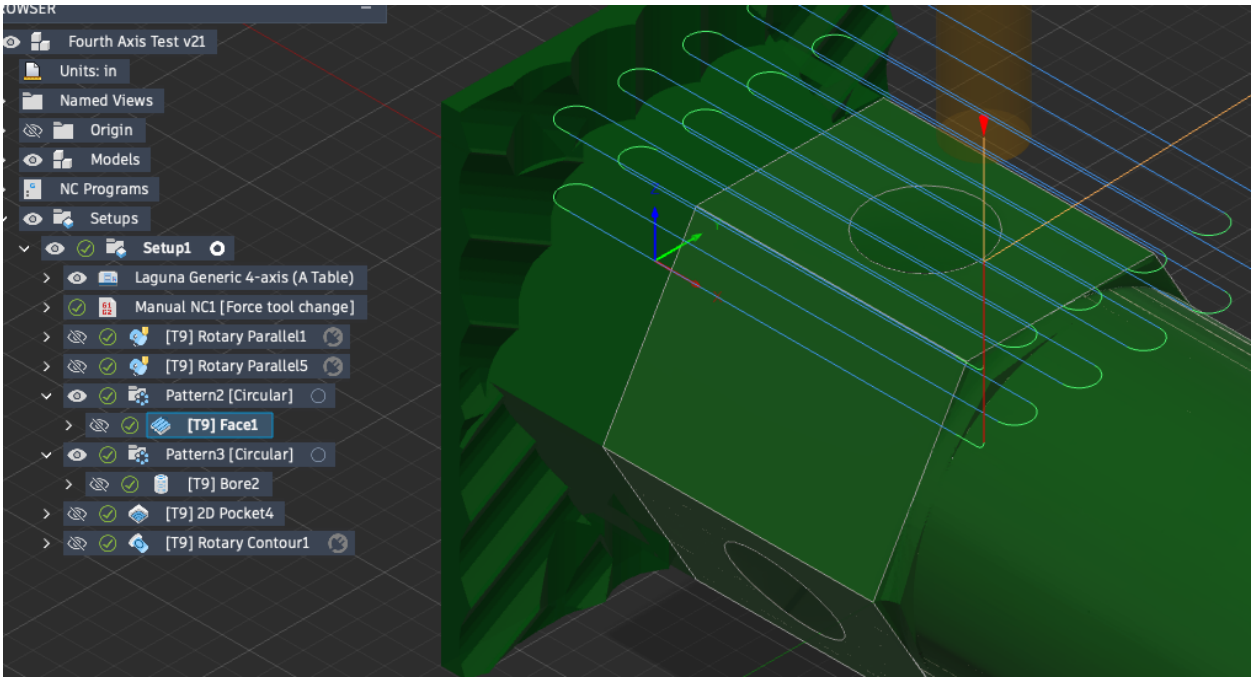


Rotary Pattern

Does not require the machining extension. This is an indexing operation, inside of which you can use 2D toolpaths. Start the pattern by selection Pattern from the Setup menu.

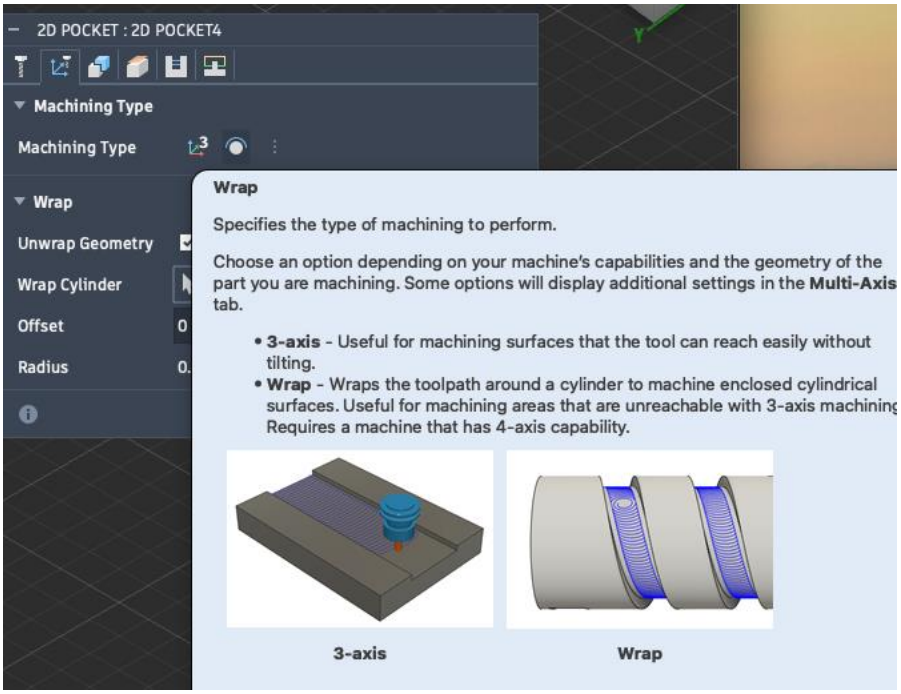


Under the pattern operation you can add 2D toolpaths”



Wrapped 2D

This takes a normal 2D toolpath and wraps it around a cylinder. A limitation is the toolpath cannot go all the way around and meet where it began. It either must be a spiral or slot of some sort. Only certain 2D operations support wrap.



Machine Setup

Position Rotary

Position the Rotary along the X axis. Use the dog holes to make sure the Rotary is aligned to the X axis. Secure using screws.

Plug in the rotary into the receptacle which is on the side of the control box close to the floor.

Rotary Controls

The rotary axis is referred to as either the “C” or “A” axis on the control. Use “C” on the control to move the rotary axis. To zero the rotary axis, Select “ZC->0” and select “A”.

Workholding

For workholding, you can turn a tenon on your stock and grab the stock in the chuck. There are internal and external chuck jaws to handle various stock sizes. There is also a faceplate, which you can secure to your workpiece using screws. If you do this, make sure you will not machine

into the area where the screws are.

Clearance

Install Tool bit and verify clearance. Make sure that the bit can clear the highest parts of the rotary. Usually, the faceplate or the belt.

Zeroing

XY Axis Zero

Align Y axis to the centerline on your part. Ideally use a toolbit with a point for accuracy. If not accurate your machined diameters will be oversized.

Move to your X axis zero on your part. When selecting this take care that the tool path will not hit the chuck, faceplate, or center.

“XY->0”

Z Axis Zero

The Z zero should be setup to be the middle of the chuck. This is 65mm from the metal base of the rotary.

First, use the Z-touch to touch off the base:

Then, move the Z axis to 65mm, and zero the Z axis.

C / A Axis Zero

Use the C buttons to rotate the axis to your desired start point, if needed. Zero by using “ZC->0” and Select the A axis.